

Challenge (Habanero)

- Q1.** A hiker consumes $2\frac{3}{4}$ liters of water per hour. How many liters will she consume in $2\frac{1}{2}$ hours?
(A) 6
(B) 6.5
(C) 7
(D) 7.5
(E) 8
- Q2.** A bird sanctuary has $15\frac{3}{8}$ acres of land. If $2\frac{1}{4}$ acres are reserved for a new species, what percent of the land is reserved?
(A) 12
(B) 13
(C) 14
(D) 15
(E) 16
- Q3.** A car travels $25\frac{1}{2}$ miles per gallon. If gas costs $2\frac{3}{4}$ per gallon, how much will it cost to drive 100 miles?
(A) 10
(A) 11
(A) 12
(A) 13
(A) 14
- Q4.** A group of friends want to split a bill of $45\frac{1}{2}$. If they split it evenly among $3\frac{1}{4}$ people, how much will each person pay?
(A) 13.50
(A) 14
(A) 14.50
(A) 15
(A) 15.50
- Q5.** A mountain has a height of $8\frac{3}{4}$ kilometers. If a hiker climbs $2\frac{1}{2}$ kilometers, what percent of the mountain has she climbed?
(A) 25
(B) 28
(C) 30
(D) 32
(E) 35
- Q6.** A park ranger measures the length of a trail to be $10\frac{1}{4}$ miles. If she measures it again and gets $10\frac{1}{2}$ miles, what is the percent increase in length?
(A) 20
(B) 25
(C) 30
(D) 35
(E) 40
- Q7.** A group of hikers consume $3\frac{3}{4}$ pounds of food per day. If they have $15\frac{1}{2}$ pounds of food, how many days will it last?
(A) 3.5
(B) 4
(C) 4.5
(D) 5
(E) 5.5
- Q8.** A car's fuel tank can hold $15\frac{1}{4}$ gallons. If $2\frac{3}{4}$ gallons are already in the tank, what percent is filled?
(A) 15
(B) 16
(C) 17
(D) 18
(E) 19

Open Ended Questions (FRQ)

- Q9.** A wildlife reserve has $20\frac{1}{2}$ acres of land for animal habitats. If $3\frac{1}{4}$ acres are for bears and $2\frac{1}{2}$ acres are for wolves, what fraction of the land is left for other animals? Show your work and explain your reasoning.
- Q10.** A hiker is planning a trip and needs to calculate her fuel consumption. Her backpack weighs $25\frac{3}{4}$ pounds and she will be hiking for $5\frac{1}{2}$ hours. If she consumes $2\frac{1}{4}$ pounds of food per hour, how many pounds of food will she consume on her trip? Additionally, if she wants to bring $1\frac{1}{2}$ pounds of extra food, what fraction of her backpack's weight will be food? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Remind students to convert between fractions, decimals, and percents where necessary.
- Tip 2: Encourage students to use number lines to visualize rational numbers.
- Tip 3: Suggest that students use estimations to check their calculations.